

MAT1002 Lecture 11,

Outline

- Vector-valued functions
- Limits and continuity
- Derivatives
- Integrals

Vector-Valued Functions (on Intervals)

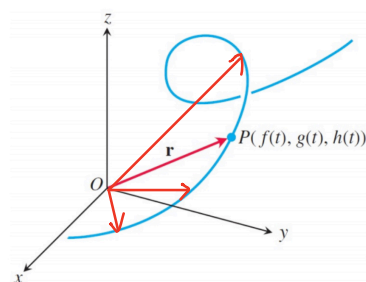
Definition

(on I)

A **vector-valued function** or a **vector function** is a function $\vec{r}: I \rightarrow \mathbb{R}^n$, where I is an interval and $\mathbb{R}^n$ is viewed as a set of vectors.

Remark

For a vector function $\vec{r}: I \rightarrow \mathbb{R}^n$, we have $\vec{r}(t) = \langle r_1(t), \dots, r_n(t) \rangle$, where each r_i is a function from I to $\mathbb{R}$.



Example

Sketch the curve parametrized by the function

$$\vec{r}(t) := \langle \cos(t), \sin(t), t \rangle, \quad t \in \mathbb{R}.$$

This curve is called a **helix**.

Limits and Continuity

Definition

Let $\vec{r}: I \rightarrow \mathbb{R}^n$ be a vector function. We say that $\vec{r}$ has a **limit** $\vec{L}$ as t approaches t_0 , and write

$$\lim_{t \rightarrow t_0} \vec{r}(t) = \vec{L},$$

if, for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all $t \in I$, $0 < |t - t_0| < \delta$ implies $|\vec{r}(t) - \vec{L}| < \epsilon$.

The following is a convenient equivalent condition.

Theorem (“Component-Wise Limits”)

A vector function $\vec{r} = \langle r_1, r_2, \dots, r_n \rangle$ satisfies

$$\lim_{t \rightarrow t_0} \vec{r}(t) = \vec{L} = \langle L_1, L_2, \dots, L_n \rangle$$

if and only if $\lim_{t \rightarrow t_0} r_i(t) = L_i$ for all i .

Example

If $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$, then

$$\begin{aligned} \lim_{t \rightarrow \pi/4} \mathbf{r}(t) &= \left(\lim_{t \rightarrow \pi/4} \cos t \right) \mathbf{i} + \left(\lim_{t \rightarrow \pi/4} \sin t \right) \mathbf{j} + \left(\lim_{t \rightarrow \pi/4} t \right) \mathbf{k} \\ &= \frac{\sqrt{2}}{2} \mathbf{i} + \frac{\sqrt{2}}{2} \mathbf{j} + \frac{\pi}{4} \mathbf{k}. \end{aligned}$$

Definition

A vector function $\vec{r}: I \rightarrow \mathbb{R}^n$ is said to be **continuous** at $t_0 \in I$ if

$$\lim_{t \rightarrow t_0} \vec{r}(t) = \vec{r}(t_0),$$

and it is called a **continuous function** if it is continuous at all points in I .

Remark By component-wise limits, it follows that $\vec{r} = \langle r_1, \dots, r_n \rangle$ is continuous at $t_0 \Leftrightarrow r_i$ is continuous at t_0 , $\forall i$.

Example

The function $\vec{r}(t) := \langle \cos(t), \sin(t), \lfloor t \rfloor \rangle$ is discontinuous at every $t \in \mathbb{Z}$, since so is the third component.

Derivatives

Definition

- ▶ A vector function $\vec{r}$ is said to be **differentiable at t** if the limit

$$\vec{r}'(t) := \frac{d\vec{r}}{dt} := \lim_{h \rightarrow 0} \frac{\vec{r}(t+h) - \vec{r}(t)}{h}$$

exists.

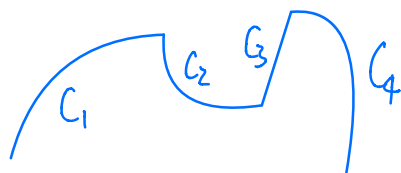
h can be thought of a Δt .

- ▶ The vector $\vec{r}'(t)$ is called the **derivative** of $\vec{r}$ at t .
- ▶ If $\vec{r}'(t) \neq \vec{0}$, $\vec{r}'(t)$ is also called the **tangent vector** to the curve (given by $\vec{r}$) at the point $P = P(t)$, where $\vec{OP} = \vec{r}(t)$.
- ▶ The corresponding **unit tangent vector** is

$$\vec{T}(t) := \frac{\vec{r}'(t)}{|\vec{r}'(t)|}$$

- ▶ The **tangent line** to the curve at a point $P(t)$ is the line through $P(t)$ in the direction of $\vec{r}'(t)$.
- A curve is said to be **smooth** if it can be traced by a vector function $\vec{r}$ s.t. $d\vec{r}/dt$ is continuous and is never $\vec{0}$ for any t .

e.g.



← Not smooth, but
piecewise smooth

← Each C_i is smooth

Remark Although usually draw in plane or 3D-space, the definitions above apply to $\vec{r}: I \rightarrow \mathbb{R}^n$ for any $n (\geq 1)$.

Physical Meanings

If $\vec{r}(t)$ is the position vector of a particle moving along a curve at time t , then:

- ▶ $\vec{r}'(t)$ is the particle's **velocity vector** at time t , and;
- ▶ $|\vec{r}'(t)|$ is the particle's **speed** at time t .

Note that

$$\text{Velocity} = \vec{r}'(t) = |\vec{r}'(t)| \left(\frac{\vec{r}'(t)}{|\vec{r}'(t)|} \right) = (\text{Speed})(\text{Direction}).$$

- $\frac{d\vec{r}'}{dt} = \frac{d(d\vec{r}/dt)}{dt} = \frac{d^2\vec{r}}{dt^2}$ is the particle's **acceleration vector**.
- $\vec{T}(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}$ is the **direction of motion** at time t .

The derivative of a vector function can be computed by taking derivatives component-wise.

Theorem

If $\vec{r}(t) := \langle r_1(t), r_2(t), \dots, r_n(t) \rangle$ is a vector function, where each r_i is differentiable, then

$$\vec{r}'(t) = \langle r_1'(t), r_2'(t), \dots, r_n'(t) \rangle.$$

This follows from component-wise limits.

e.g A curve is traced out by $\vec{r}(t) := \langle t+t^3, te^{-t}, \sin(2t) \rangle$.
Find direction of motion and acceleration of the particle at the point $(1, 0, 0)$.

Sol: $\vec{r}'(t) = \langle 3t^2, -te^{-t} + e^{-t}, 2\cos(2t) \rangle$;

$\vec{r}(t) = \langle 1, 0, 0 \rangle \Leftrightarrow t = 0$

$\Rightarrow \vec{r}'(1) = \langle 0, 1, 2 \rangle \Rightarrow \vec{r}(0) = \frac{\vec{r}'(0)}{|\vec{r}'(0)|} = \frac{1}{\sqrt{5}} \langle 0, 1, 2 \rangle$. Direction

$\frac{d\vec{r}'(t)}{dt} = \langle 6t, te^{-t} - e^{-t} - e^{-t}, -4\sin(2t) \rangle$

$= \langle 6t, (t-2)e^{-t}, -4\sin(2t) \rangle$

$\Rightarrow \left. \frac{d\vec{r}'(t)}{dt} \right|_{t=0} = \langle 0, -2, 0 \rangle$. Acceleration

Differentiation Rules

Differentiation Rules for Vector Functions

Let $\mathbf{u}$ and $\mathbf{v}$ be differentiable vector functions of t , $\mathbf{C}$ a constant vector, c any scalar, and f any differentiable scalar function.

1. Constant Function Rule: $\frac{d}{dt} \mathbf{C} = \mathbf{0}$

2. Scalar Multiple Rules: $\frac{d}{dt} [c\mathbf{u}(t)] = c\mathbf{u}'(t)$

$\frac{d}{dt} [f(t)\mathbf{u}(t)] = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t)$

3. Sum Rule: $\frac{d}{dt} [\mathbf{u}(t) + \mathbf{v}(t)] = \mathbf{u}'(t) + \mathbf{v}'(t)$

4. Difference Rule: $\frac{d}{dt} [\mathbf{u}(t) - \mathbf{v}(t)] = \mathbf{u}'(t) - \mathbf{v}'(t)$

5. Dot Product Rule: $\frac{d}{dt} [\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$

6. Cross Product Rule: $\frac{d}{dt} [\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$

7. Chain Rule: $\frac{d}{dt} [\mathbf{u}(f(t))] = f'(t)\mathbf{u}'(f(t))$

Scalar function/
real-valued function

Properties 1, 2, 3, 4 and 7 can be proven component-wise.

Property 5 can be verified directly.

To prove property 6, we use the following fact:

$$\lim_{t \rightarrow t_0} (\vec{a}(t) \times \vec{b}(t)) = \left(\lim_{t \rightarrow t_0} \vec{a}(t) \right) \times \left(\lim_{t \rightarrow t_0} \vec{b}(t) \right).$$

This fact can be proven by considering components.

Proof of Property 6:

$$\begin{aligned} \frac{d}{dt} (\vec{a}(t) \times \vec{v}(t)) &= \lim_{h \rightarrow 0} \frac{\vec{a}(t+h) \times \vec{v}(t+h) - \vec{a}(t) \times \vec{v}(t)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\vec{a}(t+h) \times \vec{v}(t+h) - \vec{a}(t+h) \times \vec{v}(t) + \vec{a}(t+h) \times \vec{v}(t) - \vec{a}(t) \times \vec{v}(t)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\vec{a}(t+h) \times (\vec{v}(t+h) - \vec{v}(t))}{h} + \lim_{h \rightarrow 0} \frac{(\vec{a}(t+h) - \vec{a}(t)) \times \vec{v}(t)}{h} \\ &= \lim_{h \rightarrow 0} \underbrace{\vec{a}(t+h)}_{\vec{a} \text{ cts since differentiable}} \times \lim_{h \rightarrow 0} \frac{(\vec{v}(t+h) - \vec{v}(t))}{h} + \lim_{h \rightarrow 0} \frac{(\vec{a}(t+h) - \vec{a}(t))}{h} \times \lim_{h \rightarrow 0} \vec{v}(t) \\ &= \vec{a}(t) \times \vec{v}'(t) + \vec{a}'(t) \times \vec{v}(t). \end{aligned}$$



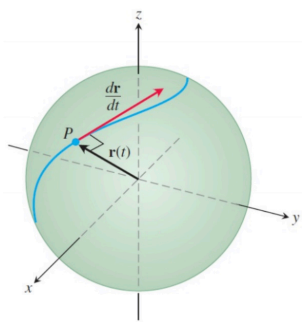
e.g. (Chain rule) If $\vec{r}(t) = \langle \sin(t^2), e^{t^2} \rangle$ then

$$\vec{r}'(t) = 2t \langle \cos(t^2), e^{t^2} \rangle.$$

Vector Functions of Constant Length

Suppose that a curve given by a differentiable function $\vec{r}$ lies on a sphere centered at the origin. Then at any t , the tangent vector $\vec{r}'(t)$ is perpendicular to the position vector $\vec{r}(t)$. This can be shown by algebraically proving the following statement: if $\vec{r}$ is a differentiable vector function of constant length, then for all t ,

$$\vec{r}(t) \cdot \vec{r}'(t) = 0.$$



Proof: $|\vec{r}(t)| \equiv a$ Constantly equal to

$$\Rightarrow \vec{r}(t) \cdot \vec{r}(t) = |\vec{r}(t)|^2 \equiv a^2$$

$$\Rightarrow \vec{r}(t) \cdot \vec{r}'(t) + \vec{r}'(t) \cdot \vec{r}(t) \equiv 0$$

Dot product rule

$$\Rightarrow 2\vec{r}(t) \cdot \vec{r}'(t) \equiv 0$$

$$\Rightarrow \vec{r}(t) \cdot \vec{r}'(t) \equiv 0$$

□

Integrals

Definition

Let $\vec{r}: I \rightarrow \mathbb{R}^n$ be a vector function, with $\vec{r} = \langle r_1, r_2, \dots, r_n \rangle$.

- ▶ A function $\vec{R}$ is called an **antiderivative** of $\vec{r}$ if $\vec{R}'(t) = \vec{r}(t)$ for all $t \in I$.
- ▶ The **indefinite integral** of $\vec{r}$, denoted by $\int \vec{r}(t) dt$, is defined to be the set of all antiderivatives of $\vec{r}$. We write

$$\int \vec{r}(t) dt = \vec{R}(t) + \vec{C},$$

where $\vec{C}$ represents any constant vector.

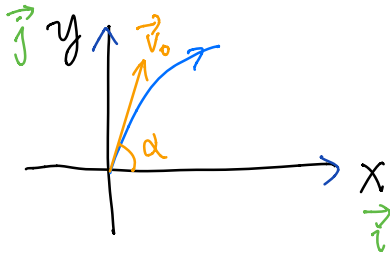
- ▶ The **definite integral**, denoted by $\int_a^b \vec{r}(t) dt$, is defined by

$$\int_a^b \vec{r}(t) dt := \left\langle \int_a^b r_1(t) dt, \int_a^b r_2(t) dt, \dots, \int_a^b r_n(t) dt \right\rangle.$$

Since integrals are defined component-wise, the FTC also works for vector functions: if $\vec{R}$ is an antiderivative of $\vec{r}$ on $[a, b]$, then

$$\int_a^b \vec{r}(t) dt = \vec{R}(b) - \vec{R}(a).$$

Application: Ideal Projectile of Motions



- Object is fired at origin at angle α .
- Assume only constant force $\vec{F}$ of gravity is applied.

$\vec{a} := \vec{a}(t)$ Acceleration

• Newton's Second Law of Motion $\Rightarrow \vec{F} = m\vec{a}$. ①

• Also, $\vec{F} = mg(-\vec{j})$ ②, where g is acceleration due to gravity,

So ①, ② $\Rightarrow \vec{a} = -g\vec{j}$

Velocity $\Rightarrow \vec{v}(t) = \int \vec{a}(t) dt = -g\vec{j}t + \vec{c}$

$\therefore \vec{v}(0) = \vec{v}_0 \quad \therefore \vec{v}_0 = \vec{c} \quad \therefore \vec{v}(t) = -gt\vec{j} + \vec{v}_0$

• $\vec{r}(t) = \int \vec{v}(t) dt = -\frac{1}{2}gt^2\vec{j} + t\vec{v}_0$

$\therefore \vec{r}(0) = \vec{0} \quad \therefore \vec{r} = \vec{0} \quad \therefore \vec{r}(t) = -\frac{1}{2}gt^2\vec{j} + t\vec{v}_0$ ③

Let $v_0 := |\vec{v}_0|$ be the initial speed. Then

$$\vec{v}_0 = v_0 \cos \alpha \vec{i} + v_0 \sin \alpha \vec{j}.$$

Substituting this into ③ yields

$$\vec{r}(t) = v_0 (\cos \alpha) t \vec{i} + (v_0 (\sin \alpha) t - \frac{1}{2} g t^2) \vec{j}.$$

This is the position vector of the object at time t . The

parametric equations are

$$x = v_0(\cos\alpha)t, \quad y = v_0(\sin\alpha)t - \frac{1}{2}gt^2, \quad t \geq 0.$$

By substituting $t = \frac{x}{v_0 \cos\alpha}$ from the left to the one on the right (assuming $\alpha \neq \pi/2$), we get

$$y = \tan\alpha x - \frac{1}{2}g \frac{1}{v_0^2 \cos^2\alpha} x^2.$$

This shows that any ideal projectile moves along a parabola. 